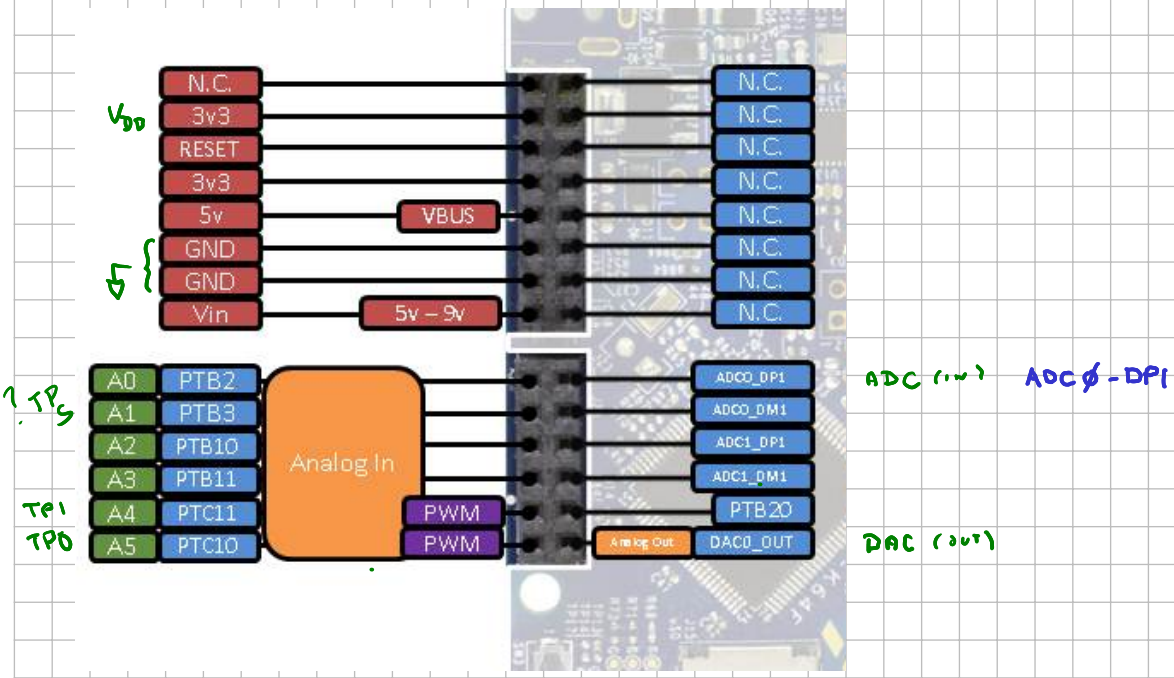
Digital Filters on K64

ADC - DAC (pin selection)



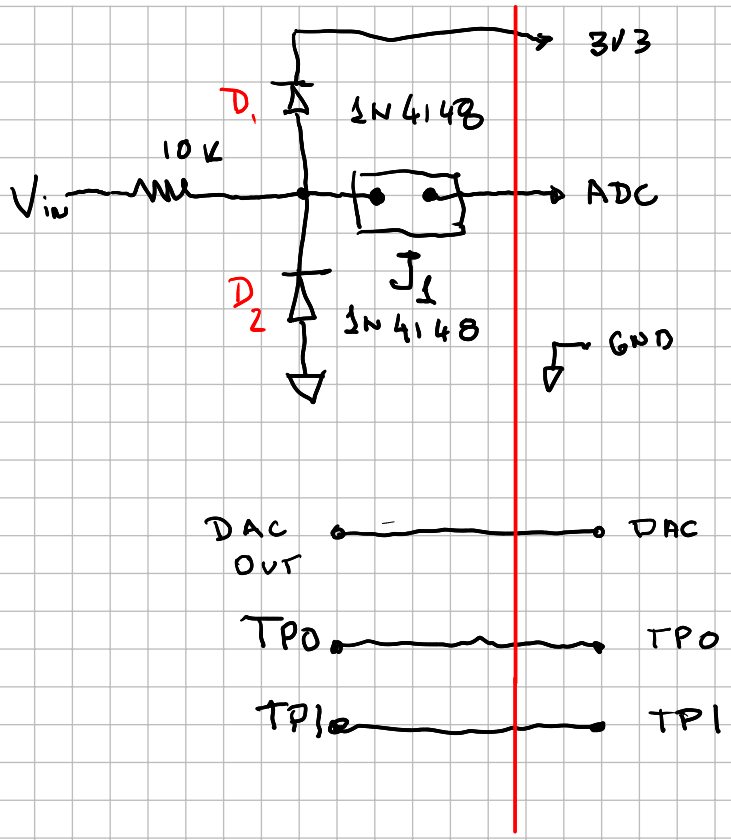
ADC0 - DP1

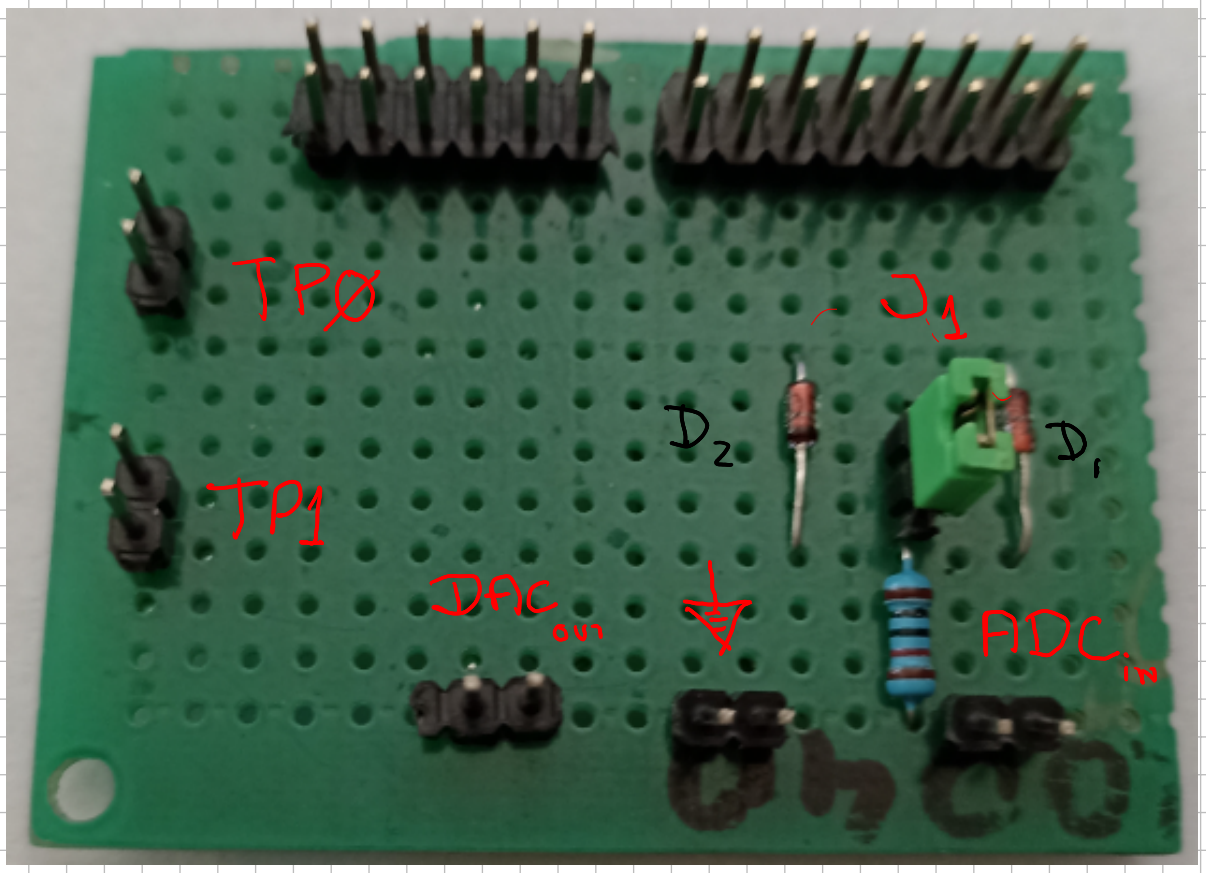


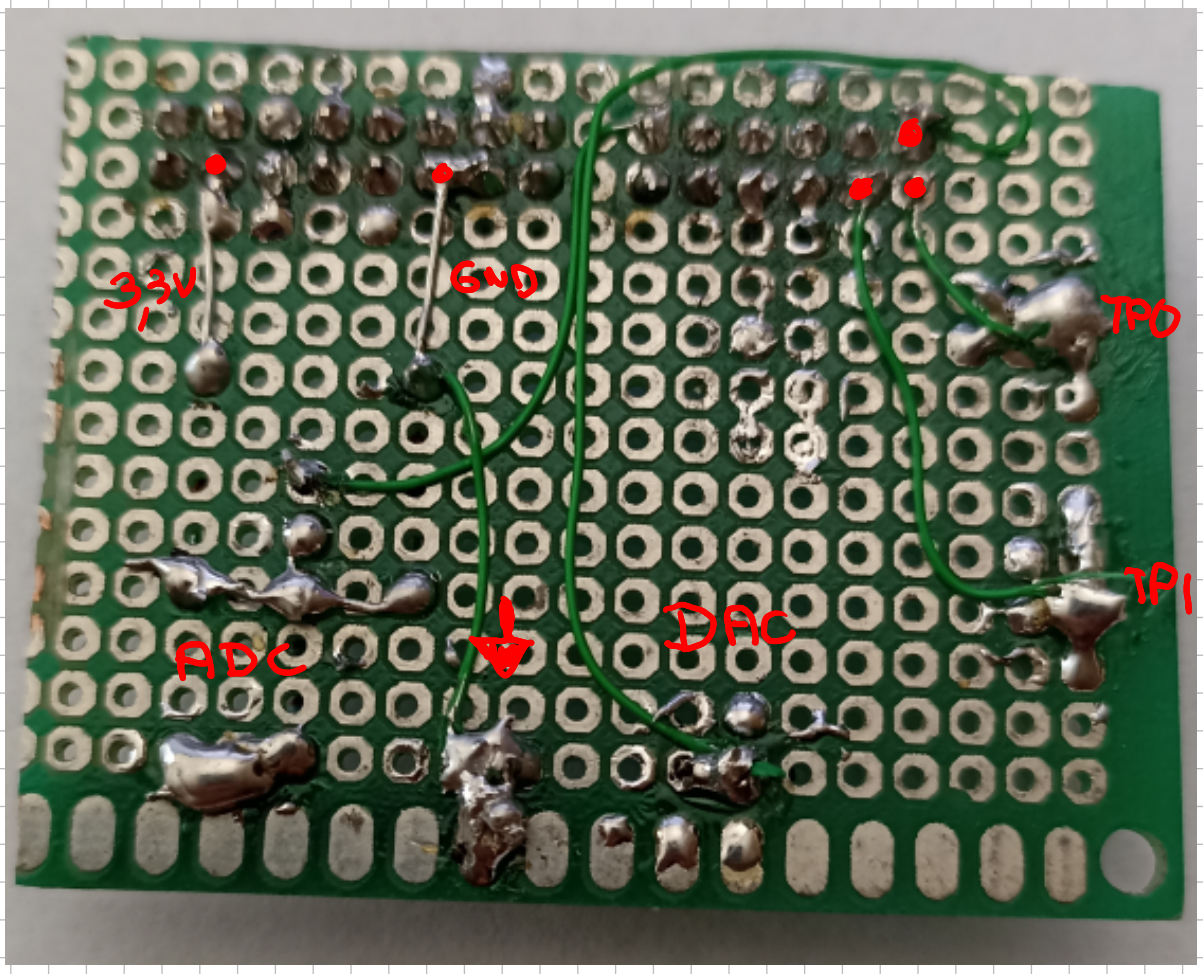
144 LQFP	144 MAP BGA	121 XFB GA	100 LQFP	Pin Name	Default	ALT0
23	J1	H1	14	ADC0_DP1	ADC0_DP1	ADC0_DP1

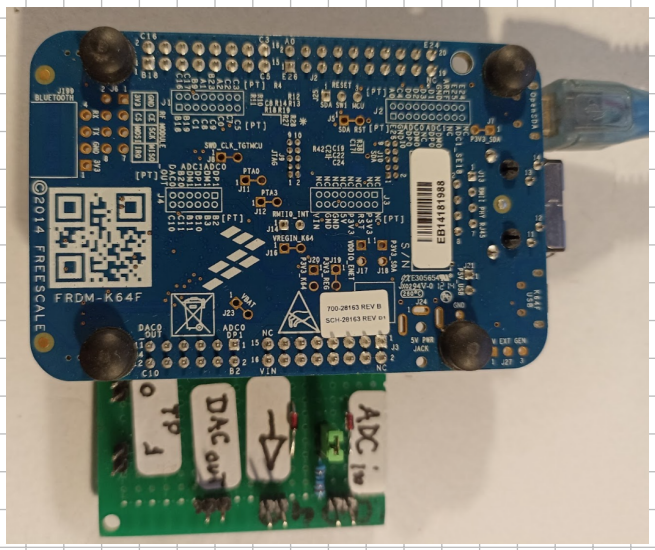
3.7.1.3.1.1 ADC0 Channel Assignment for 144-Pin Package

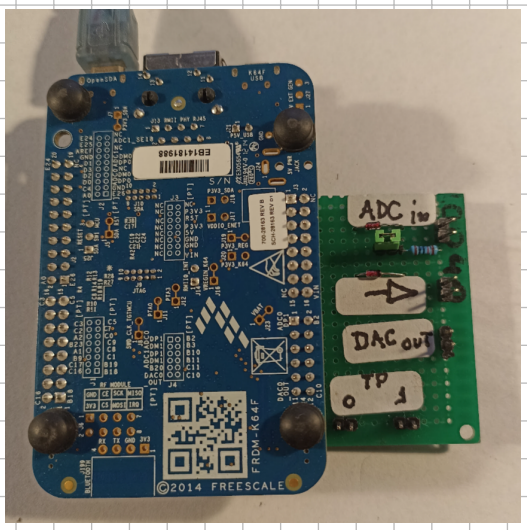
ADC Channel (SC1n[ADCH])	Channel	Input signal (SC1n[DIFF]= 1)	Input signal (SC1n[DIFF]= 0)
00000	DAD0	ADC0_DP0 and ADC0_DM0[1]	ADC0_DP0[2]
00001	DAD1	ADC0_DP1 and ADC0_DM1[3]	ADC0_DP1
00010	DAD2	ADC0_DP2 and ADC0_DM2	ADC0_DP2
00011	DAD3	ADC0_DP3 and ADC0_DM3[4]	ADC0_DP3[5]
00100[6]	AD4a	Reserved	Reserved

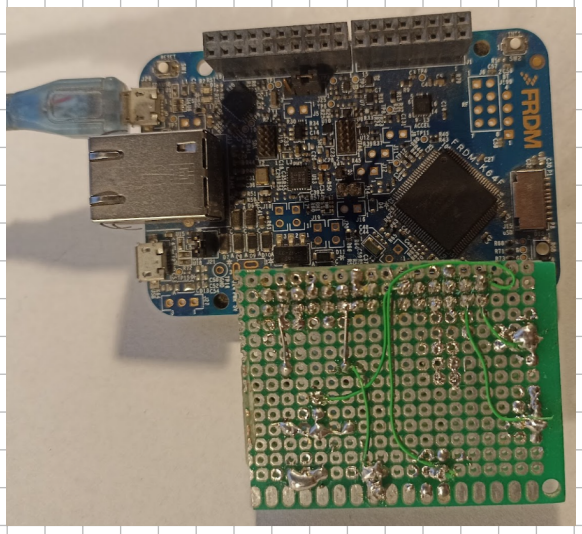

3V3
D1
1N4148
10K
Vin
ADC
J1
D2
1N4148
GND
DAC OUT
DAC
TP0
TP0
TP1
TP1


TP0
TP1
J1
D2
D1
DAC OUT
ADC IN

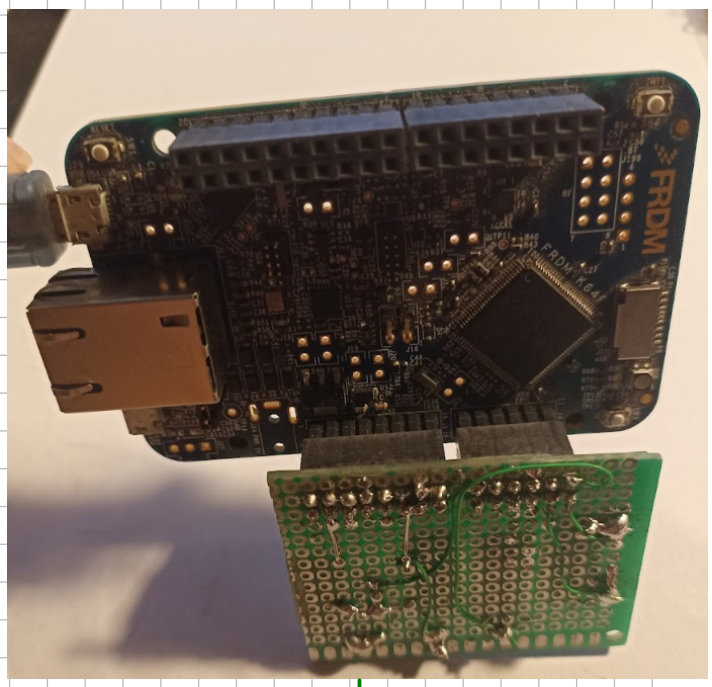

3.3V
GND
TP0
TP1
ADC
DAC

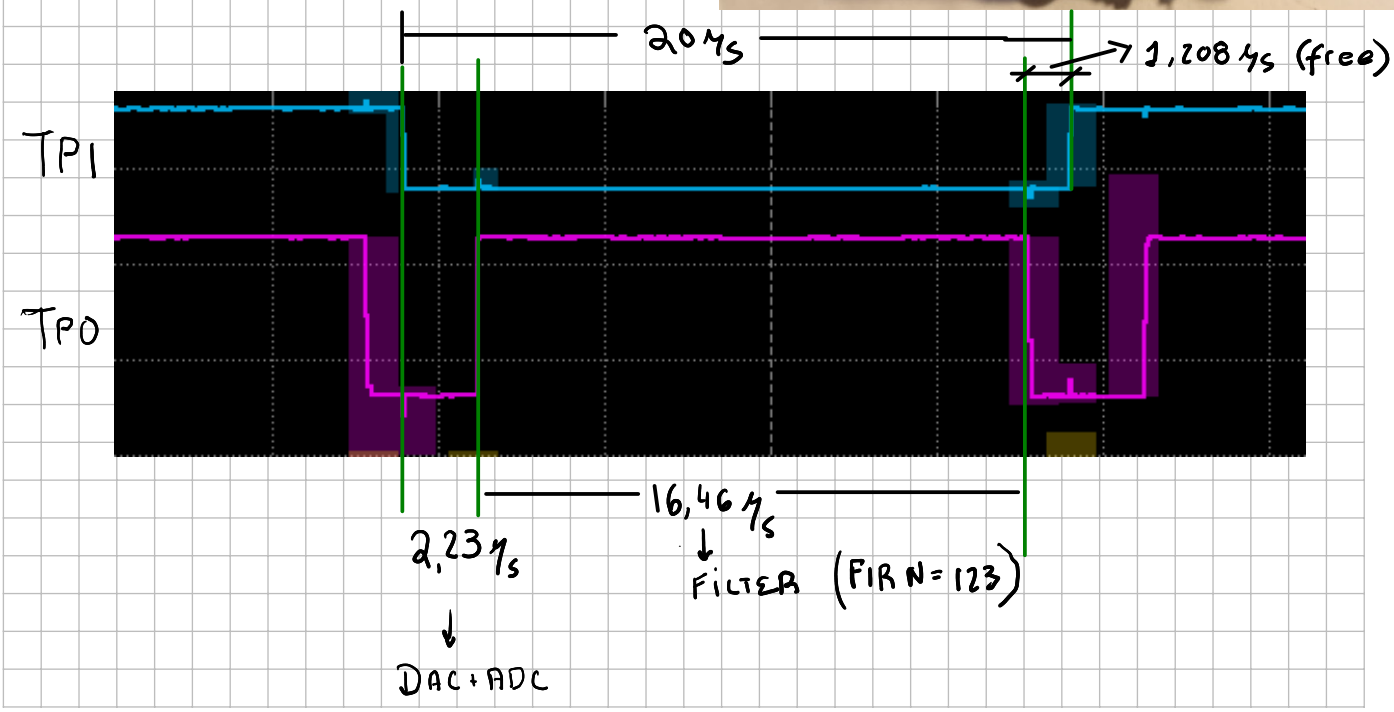

FRDM-K64F
©2014 FREESCALE
TP1
DAC out
ADC in


ADC in
DAC out
TP1
FRDM-K64F
©2014 FREESCALE

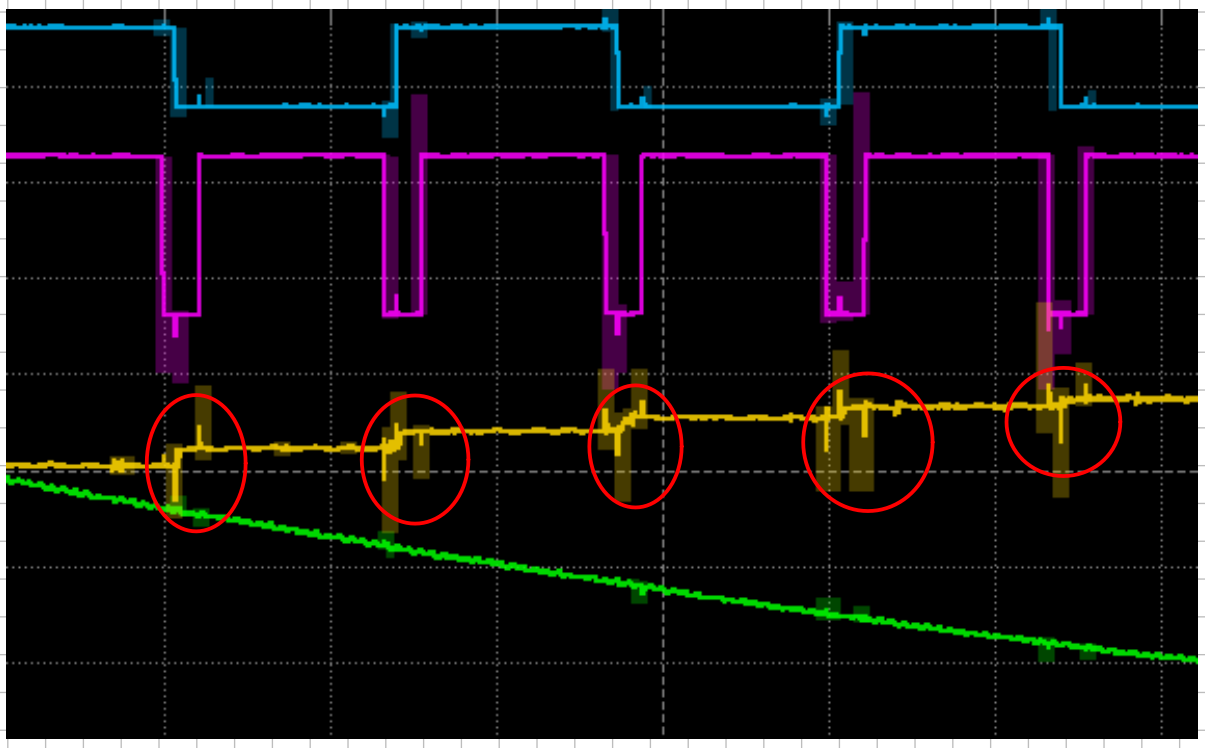



ADC
DAC out

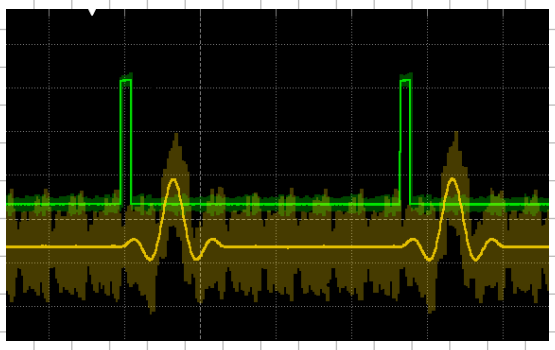



20 µs
1,208 µs (free)
TP1
TP0
16,46 µs
2,23 µs
FILTER (FIR N=123)
DAC+ADC

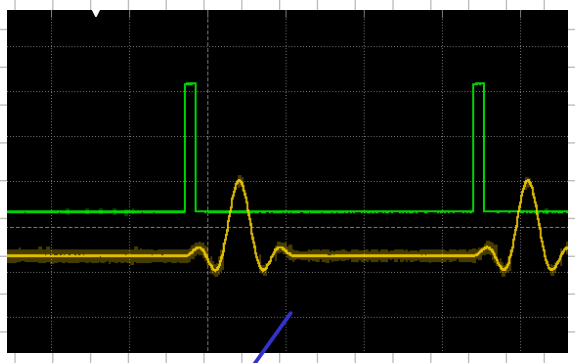
TP0, TP1 Noise

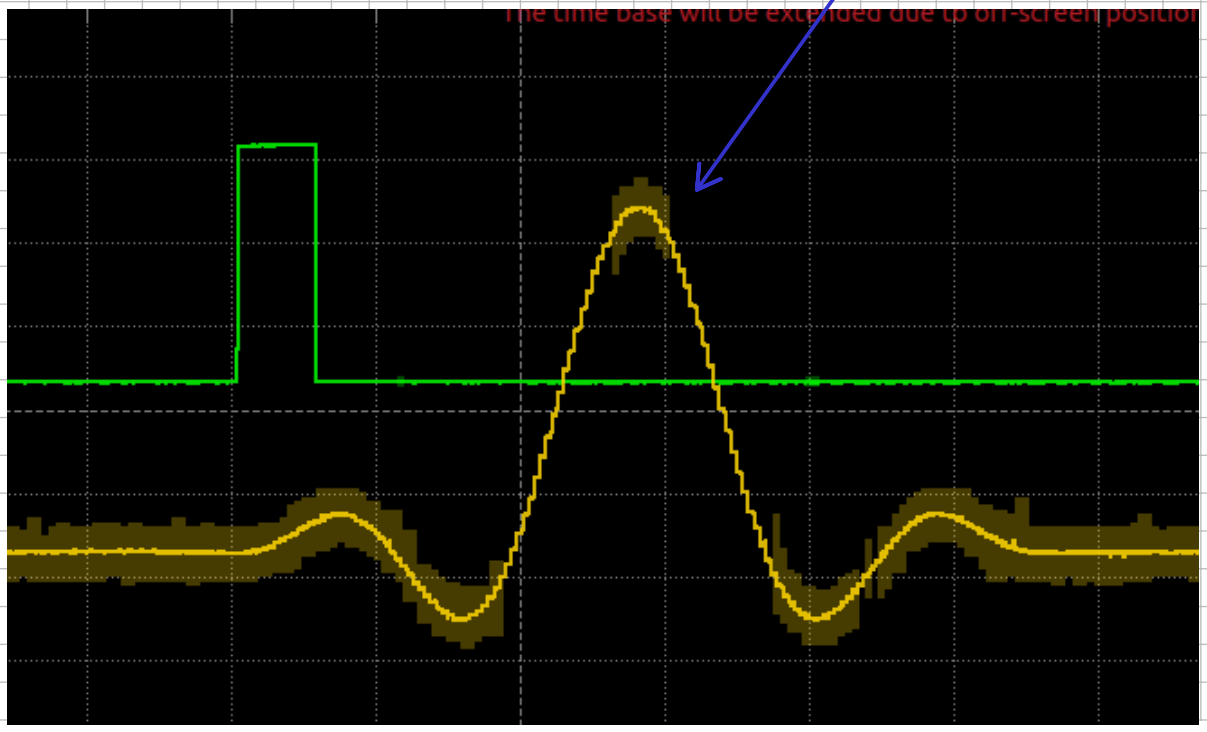


FIR impulse response



FIR impulse response





IIR impulse response

